

One CNN training step: PyTorch source and four ported candidates. Losses count as matching within 0.02.

| ported program                                 | runs to completion? | loss     | gradient norm | parameter update   | verdict from runs + loss |
|------------------------------------------------|---------------------|----------|---------------|--------------------|--------------------------|
| <b>Source program</b><br>the reference         | yes                 | 1.780532 | 0.283368      | reference          | --                       |
| <b>Correct port</b><br>no fault                | yes                 | 1.780535 | 0.283368      | matches            | accept                   |
| <b>Missing operation</b><br>no implementation  | no                  | --       | --            | --                 | reject                   |
| <b>Wrong reduction</b><br>averaged, not summed | yes                 | 2.500882 | 0.687914      | differs            | reject                   |
| <b>No backward pass</b><br>no gradients        | yes                 | 1.780535 | --            | <b>never moves</b> | <b>accept</b>            |

A completion-and-loss check accepts both shaded rows, even though their gradients and updates differ.

One of the two never updates its parameters, and no single verdict tells them apart.